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Exploration of metabolism in microbial ecosystems

From the metabolic network to the (discrete) model

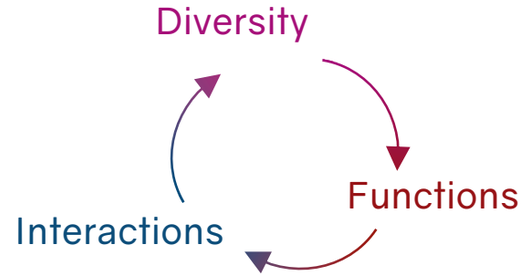
AI for Microbiome Workshop

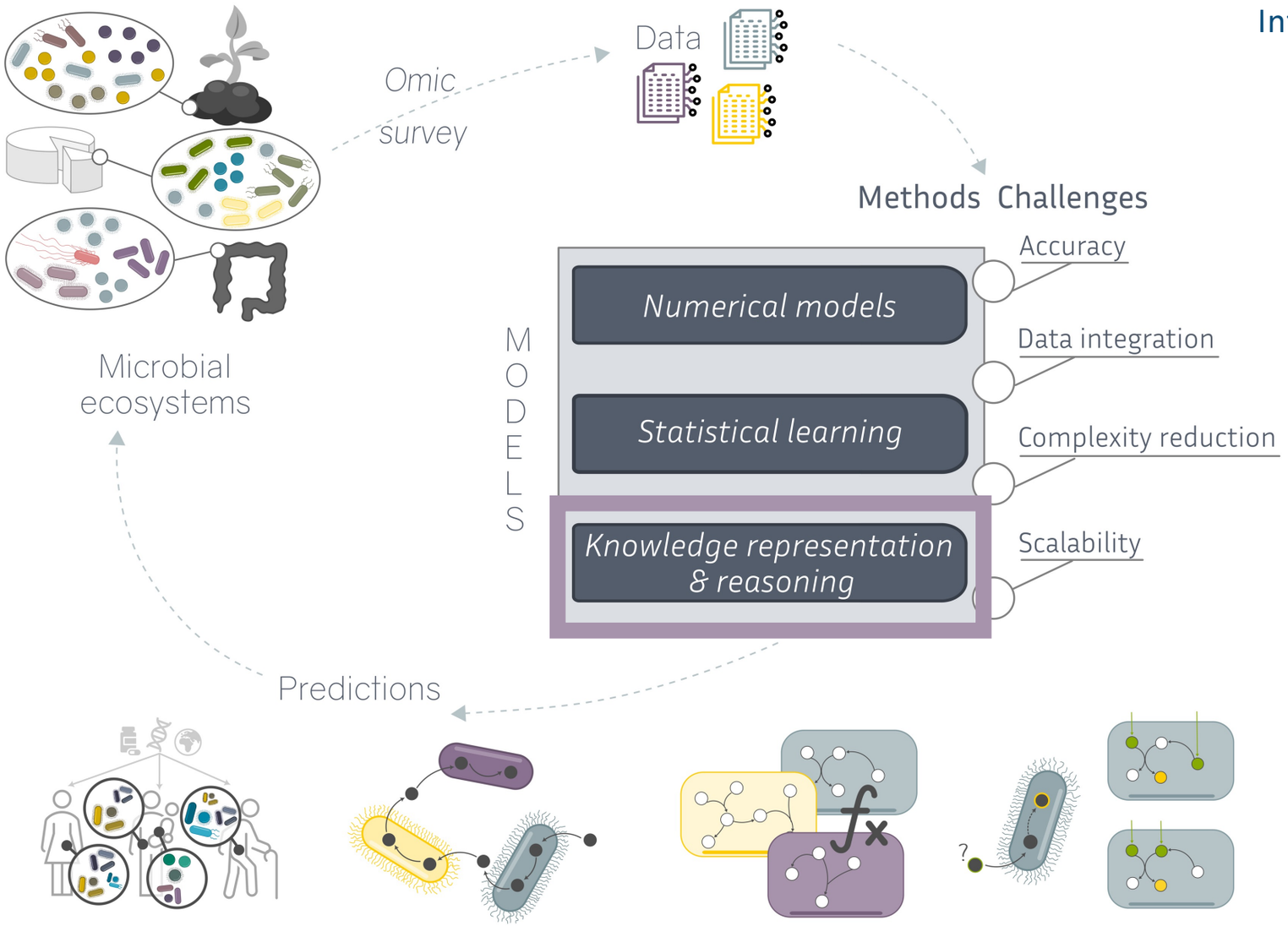
October 2nd, 2025

Inria centre at the University of Bordeaux, France

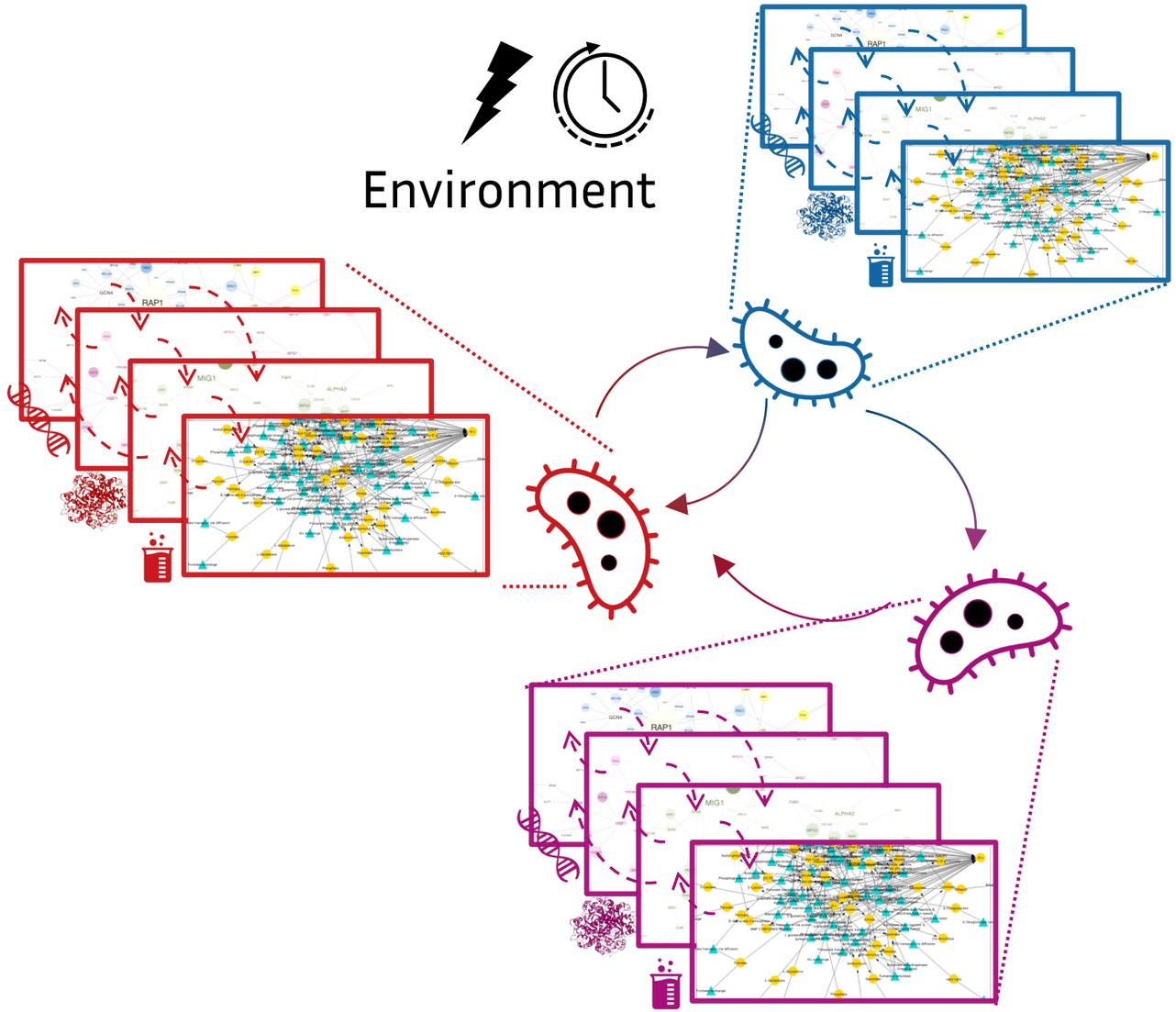
clemencefrioux@mamot.fr

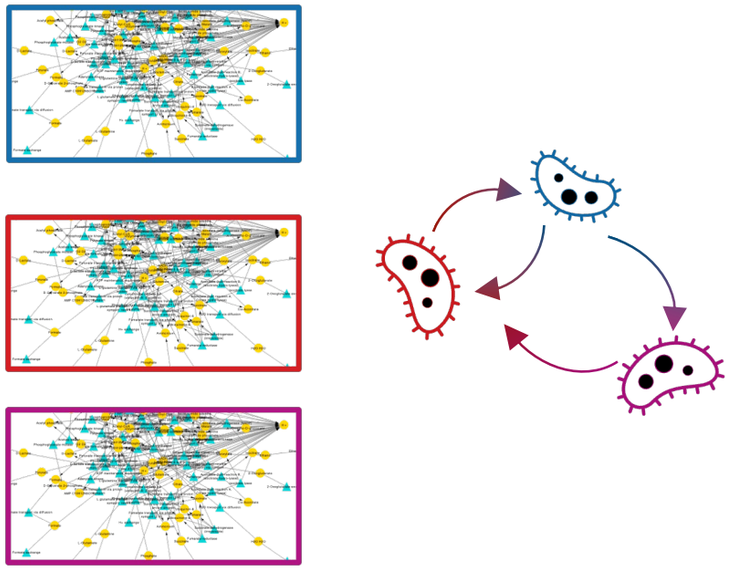
Modelling microbial ecosystems' diversity and functions





From (microbial) systems biology to systems ecology





Many interactions are mediated by metabolism

RESEARCH ARTICLE

Predicting gastrointestinal drug effects using contextualized metabolic models

Marouen Ben Guebila[1], **Ines Thiele**[1,2,3]*

1 Luxembourg Centre for Systems Biomedicine, University of Luxembourg, Esch-sur-Alzette, Luxembourg, **2** School of Medicine, National University of Ireland, Galway, University Road, Galway, Ireland, **3** Discipline of Microbiology, School of Natural Sciences, National University of Ireland, Galway, University Road, Galway, Ireland

Genome-scale metabolic reconstruction of 7,302 human microorganisms for personalized medicine

Received: 9 November 2021

Accepted: 30 November 2022

Published online: 19 January 2023

Almut Heinken[1,2,3], Johannes Hertel[1,4], Geeta Acharya[5], Dmitry A. Ravcheev[1,2], Malgorzata Nyga[6], Onyedika Emmanuel Okpala[7], Marcus Hogan[1,2], Stefanía Magnúsdóttir[8], Filippo Martinelli[1,2], Bram Nap[1,2], German Preciat[9], Janaka N. Edirisinghe[10,11], Christopher S. Henry[11], Ronan M. T. Fleming[1,9] & Ines Thiele[1,2,12,13]

Personalized whole-body models integrate metabolism, physiology, and the gut microbiome

Ines Thiele[1,2,3,4,*], Swagatika Sahoo[4,†], Almut Heinken[1], Johannes Hertel[1,5], Laurent Heirendt[4], Maike K Aurich[4] & Ronan MT Fleming[1,6,**]

Metabolic dependencies govern microbial syntrophies during methanogenesis in an anaerobic digestion ecosystem

Xinyu Zhu[1,2], Stefano Campanaro[3,4], Laura Treu[1,3*], Rekha Seshadri[2], Natalia Ivanova[2], Panagiotis G. Kougias[1,5*], Nikos Kyrpides[2] and Irini Angelidaki[1]

Dynamic co-culture metabolic models reveal the fermentation dynamics, metabolic capacities and interplays of cheese starter cultures

Emrah Özcan[1,2] | **Merve Seven**[3] | **Burcu Şirin**[3] | **Tunahan Çakır**[4] | **Emrah Nikerel**[3] | **Bas Teusink**[1] | **Ebru Toksoy Öner**[2]

Revealing the dynamics and mechanisms of bacterial interactions in cheese production with metabolic modelling

Maxime Lecomte[1,2], Wenfan Cao[1], Julie Aubert[3], David James Sherman[2], Hélène Falentin*[1], Clémence Frioux*[2], and Simon Labarthe*[4,2]

Revealing metabolic mechanisms of interaction in the anaerobic digestion microbiome by flux balance analysis

Arianna Basile[a], Stefano Campanaro[a,b,*], Adam Kovalovszki[c], Guido Zampieri[a,d], Alessandro Rossi[a], Irini Angelidaki[c], Giorgio Valle[a,1], Laura Treu[a,1]

Systematic assessment of secondary bile acid metabolism in gut microbes reveals distinct metabolic capabilities in inflammatory bowel disease

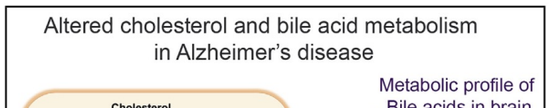
Almut Heinken[1], Dmitry A. Ravcheev[1], Federico Baldini[2], Laurent Heirendt[2], Ronan M. T. Fleming[3] and Ines Thiele[1,2,4*]

Longitudinal flux balance analyses of a patient with episodic colonic inflammation reveals microbiome metabolic dynamics

Arianna Basile[a,b], Almut Heinken[b,c], Johannes Hertel[d], Larry Smarr[e], Weizhong Li[f], Laura Treu[a], Giorgio Valle[a], Stefano Campanaro[a§], and Ines Thiele[b,c,g,h§]

Metabolic Network Analysis Reveals Altered Bile Acid Synthesis and Metabolism in Alzheimer's Disease

Graphical Abstract



Authors

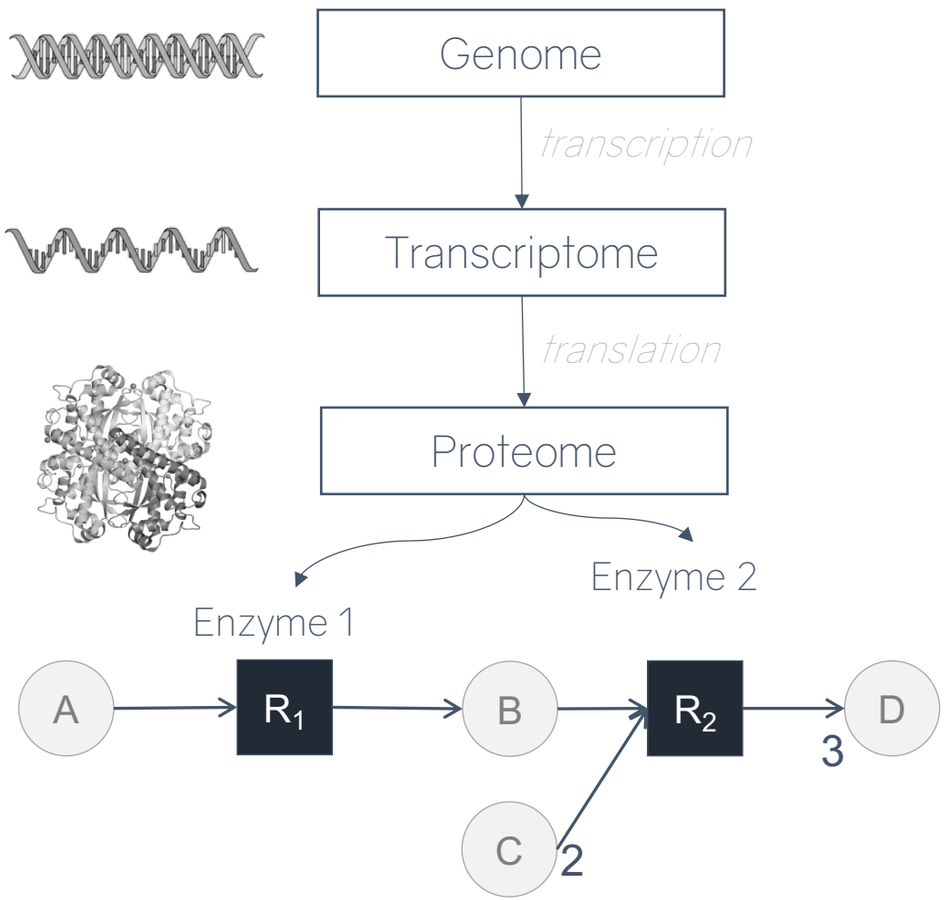
Priyanka Baloni, Cory C. Funk, Jingwen Yan, ..., The Alzheimer's Disease Metabolomics Consortium, Rima Kaddurah-Daouk, Nathan D. Price

Integrating metagenomic binning with flux balance analysis to unravel syntrophies in anaerobic CO_2 methanation

Nicola De Bernardini[1], Arianna Basile[1], Guido Zampieri[1], Adam Kovalovszki[2], Beatrix De Diego Diaz[3], Elisabetta Offer[1], Nantharat Wongfaed[4], Irini Angelidaki[5], Panagiotis G. Kougias[6*], Stefano Campanaro[1,7*†] and Laura Treu[1†]

1. Metabolic networks

Connecting genomes to functions

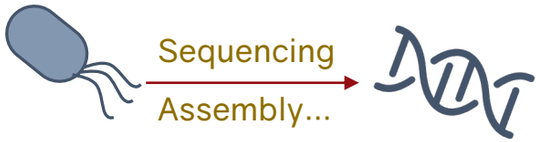


1 genome → 1,000 to 5,000 reactions

Abstraction of genomic information in genome-scale metabolic networks (GSMNs)

Functional repertoire of the cell

Metabolic networks



Sequencing

Assembly...

```
> protein_1
ATGGAATCGCATAGCA...
> protein_2
AATGCCTGATCGAATA...
```

Functional annotation

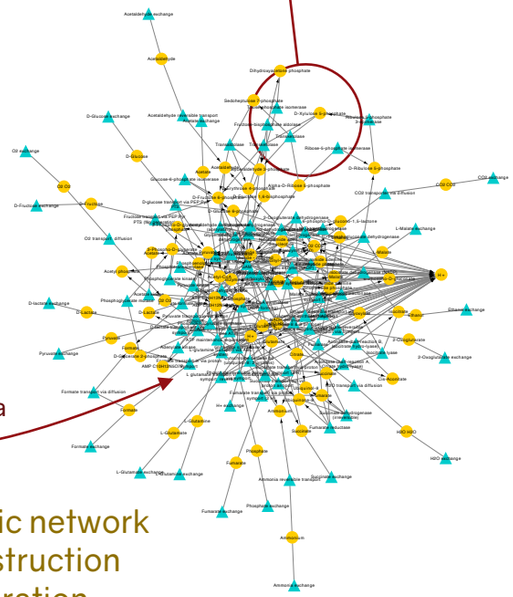
protein	name	EC-number	GO	KEGG	...
protein_1	pfkA	2.7.1.11	-	K00850	...
protein_2	eno	4.2.1.11	-	K01689	...
...	...	...	...	...	...

± other omics data

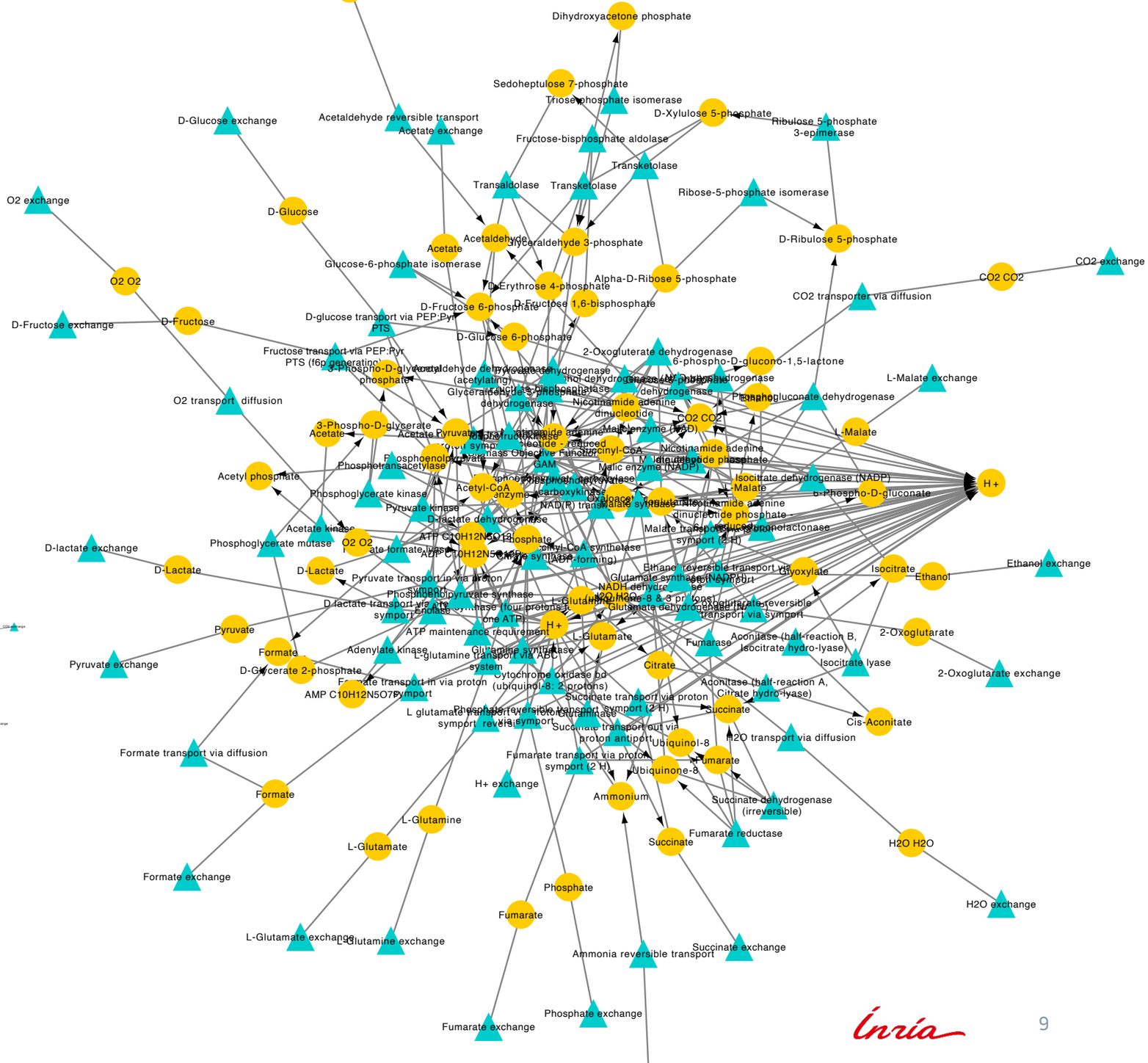
Function	
Carbohydrate degradation	
Lipid synthesis	
Vitamin biosynthesis	
Antibiotic resistance	
...	...

1000-5000 reactions

A + 2B → 2C + D

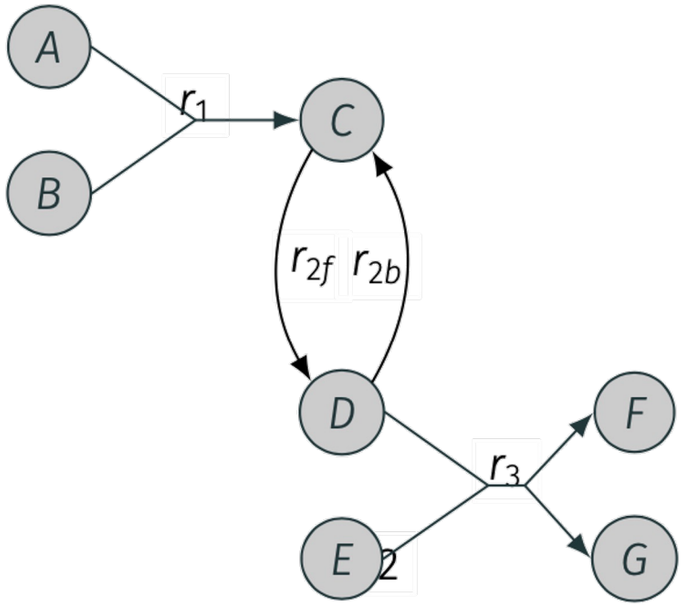


Metabolic network reconstruction & curation

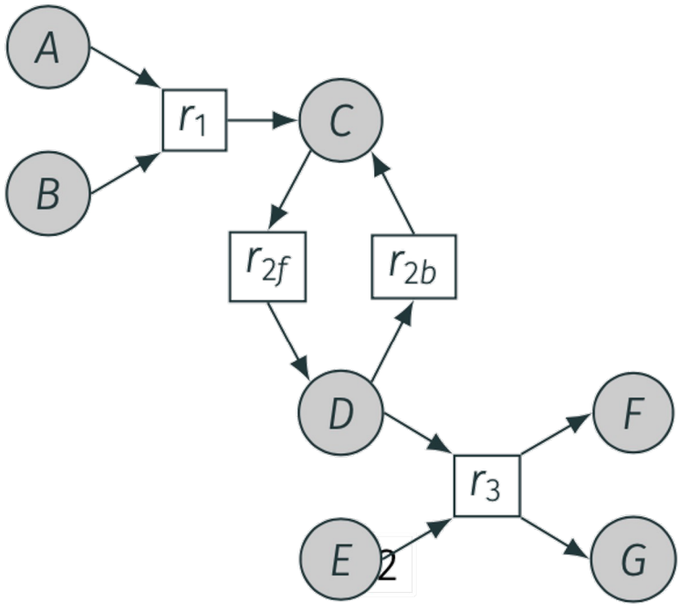


Representations of metabolic networks

$$
\begin{aligned}
r_1&: \quad A + B \rightarrow C \\
r_2&: \quad \mathrm{C} \leftrightarrow \mathrm{D} \\
r_3&: \quad D + 2E \rightarrow F + G
\end{aligned}
$$



Hypergraph



Bipartite graph

$$
S = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \\ F \\ G \end{array}
\begin{array}{c}
\begin{array}{cccc} r_1 & r_{2f} & r_{2b} & r_3 \end{array} \\
\begin{pmatrix}
-1 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 \\
+1 & -1 & +1 & 0 \\
0 & +1 & -1 & -1 \\
0 & 0 & 0 & -2 \\
0 & 0 & 0 & +1 \\
0 & 0 & 0 & +1
\end{pmatrix}
\end{array}
$$

Stoichiometric matrix

Tutorial content

https://gitlab.inria.fr/cfrioux/ebame

1. **Metabolic network exploration:**
    a. Visualisation of metabolic networks
    b. Exploration of the exchange format: SBML
2. Metabolic network reconstruction
    a. Structural annotation with Prodigal
    b. Reconstruction with gapseq
    c. Reconstruction from Kegg knowledge base
    d. Exploration of metabolic networks reconstructed with Pathway Tools
3. Metabolic network modelling of a unique species
    a. Using toy data
    b. Using real data
4. Metabolic network modelling of a community
    a. Using toy data
    b. Using real data

2. Reconstruction of metabolic networks

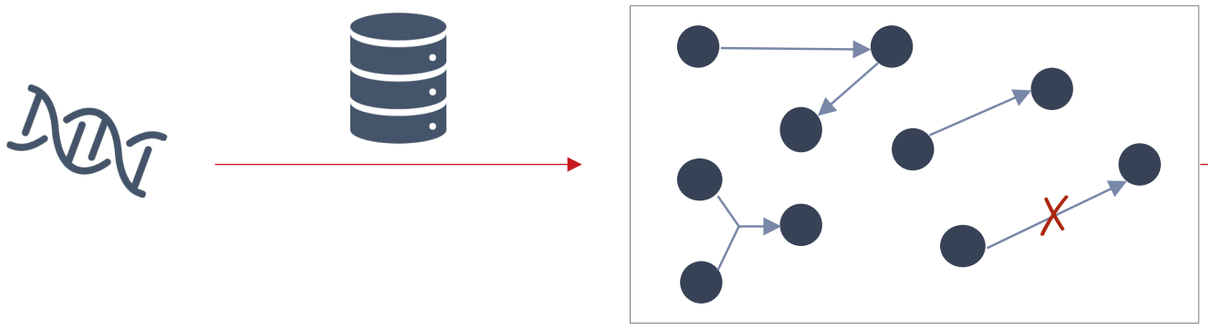
Input data

- **minimal input: a genome or a proteome**
    - a reference genome
    - a Metagenomic-Assembled Genome (MAG)
- **Genomes must be structurally annotated**
    - e.g. with Prodigal [Hyatt et al 2010] for prokaryotes
- **Some tools require also a functional annotation**
    - e.g.with EggNOG mapper [Cantalapiedra et al 2021]
    - Prokka [Seeman 2014]
    - Bakta [Schwengers 2021]

⚠ MAG completeness (Eisenhofer et al 2023)

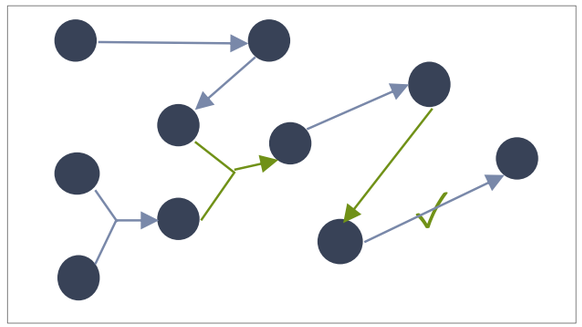
Metabolic network inference: bottom-up procedure

1. Automatic inference from sequences: *non-guided*



Gapseq [Zimmermann et al 2021]
Pathway Tools [Karp et al 2018]…

2. Semi-manual / automatic curation: *guided*

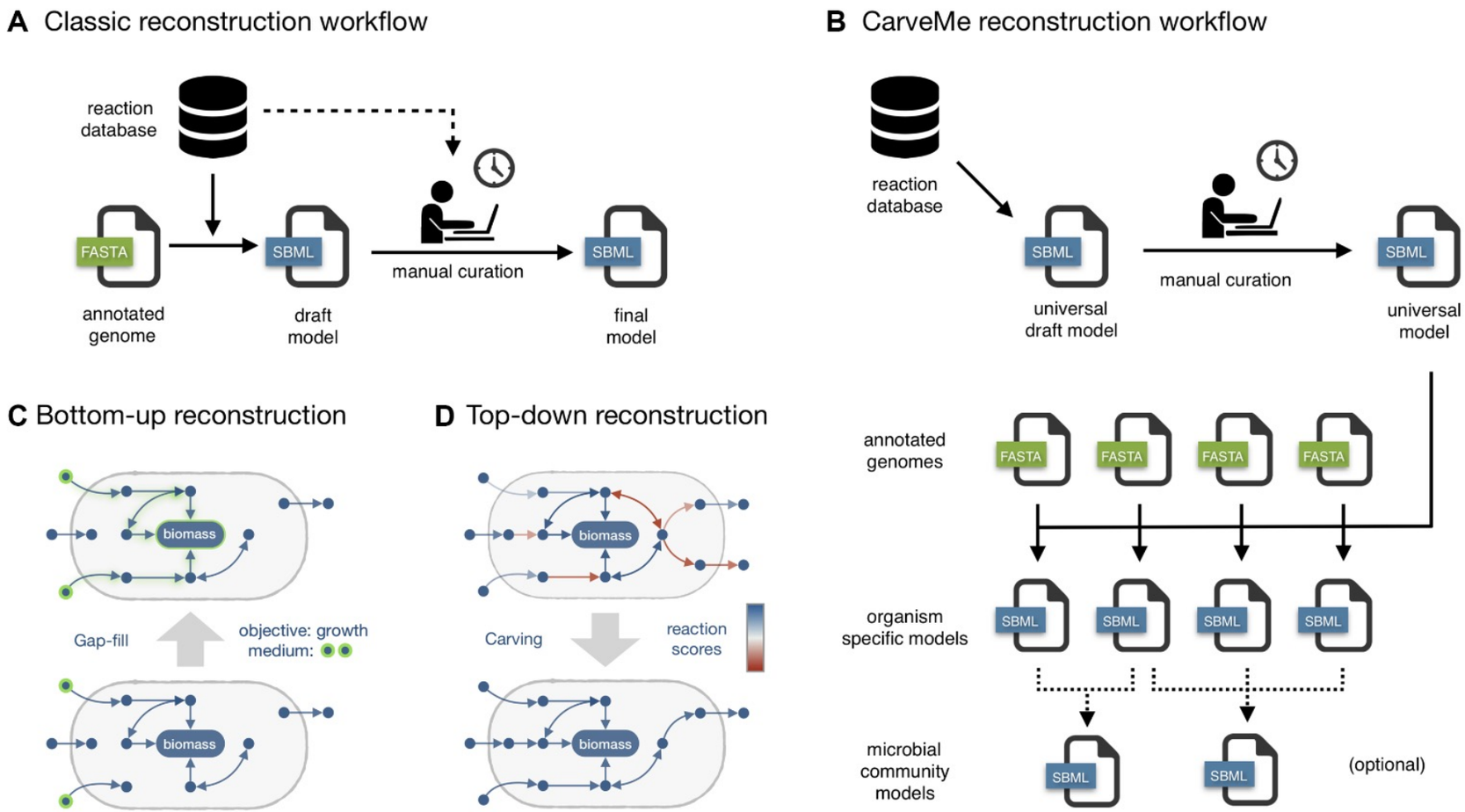


Metabolic objective
e.g. biomass / growth

Automated in gapseq [Zimmermann et al 2021]
Meneco [Prigent et al 2017],
BioISO [Cruz et al 2021]…

Can be fastidious but often necessary for high-quality GSMNs…

Alternative: top-down reconstruction



[figure from Machado et al 2018]

Metabolic models ready for constraint-based simulation

A rise in non-optimal use-cases

Cheaper sequencing
(Meta)genomics

Non-model organisms
- Limited experimentations
- Unknown biomass composition (metabolic objective)



Large-scale community modelling
- Impossible manual curation
- Uncertainty on/missing functions
- Scalability of optimisations-based simulation



→ The choice of the mathematical/computational model for simulation must account for the data and the question to be answered

Resources for (draft) reconstruction

Functional annotation

- eggNOG mapper [Cantalapiedra et al 2021]
- Prokka [Seeman 2014]
- Bakta [Schwengers 2021]

Tools scalable to microbial communities

- Pathway Tools [Karp et al 2019]
- Carveme [Machado et al 2018]
- gapseq [Zimmermann et al 2021]
- AutoKEGGRec [Karlsen et al 2018]

Collections of reconstructions

- KEGG [Kanehisa et al 2021]
- BioCyc [Karp et al 2019]
- Bigg [King et al 2016]
- Biomodels [Malik-Sheriff et al 2019]
- Virtual metabolic human [Noronha et al 2018]
- AGORA 2 [Heinken et al 2023]
- EMBL GEMs [Machado et al 2018]

A network, then what?

Exploration

- reactions, metabolic pathways, association of genes to reactions
- visualisation (usually complicated)

Modelling

- predicting the behaviour of the microbe/community in given conditions
- finding members of interest within the community
- screen the metabolic potential

Tutorial content

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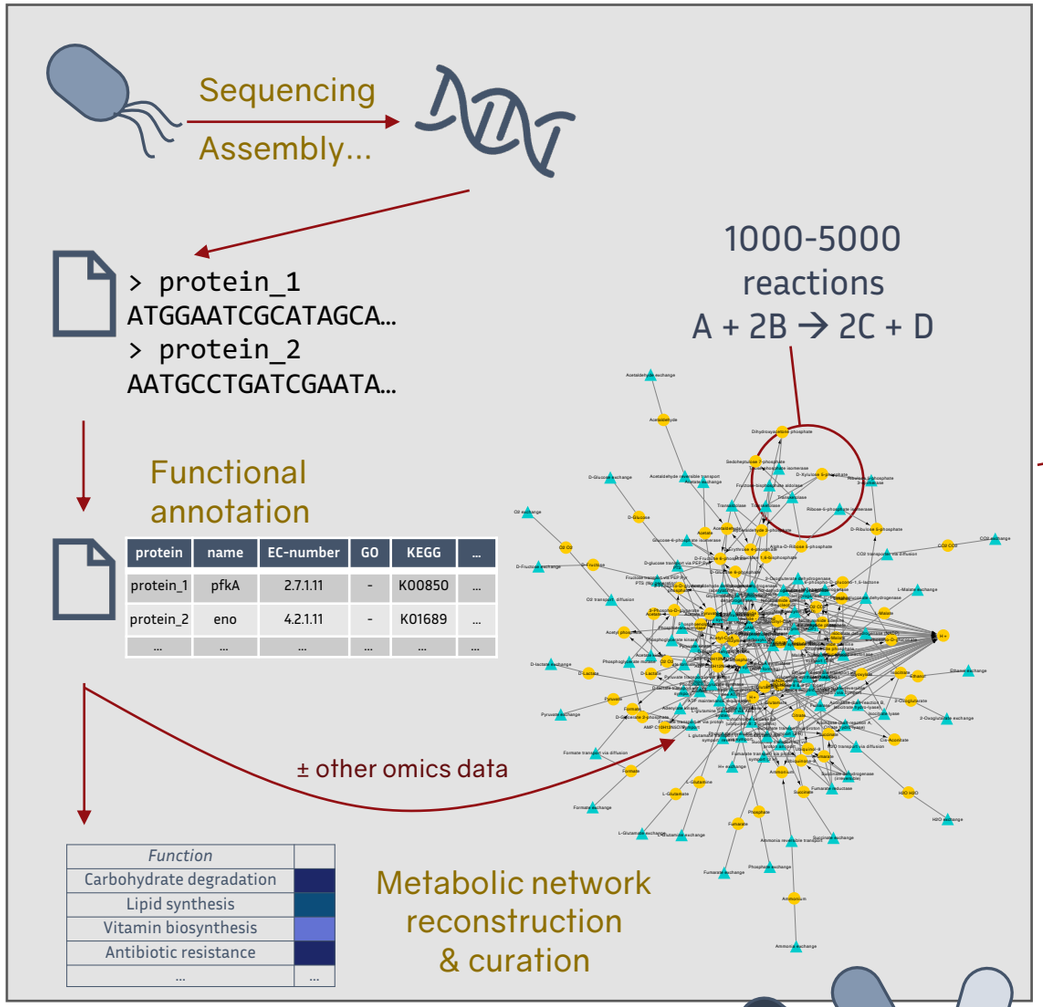
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3. Discrete metabolic models

“We were promised AI…”

Yes, but old-fashioned AI! Buckle up to enter the fascinating world of machine… reasoning

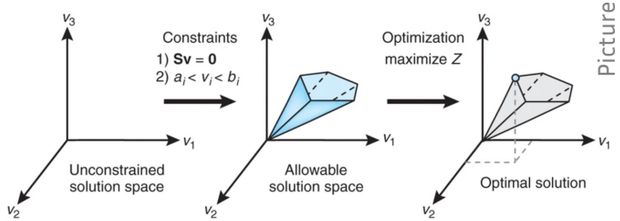
Metabolic modelling



Environmental conditions

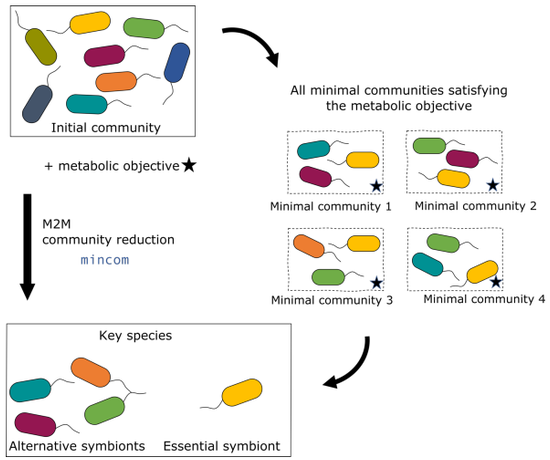
± metabolic objective

Flux analysis (FBA, FVA...)
Constraint-based modelling



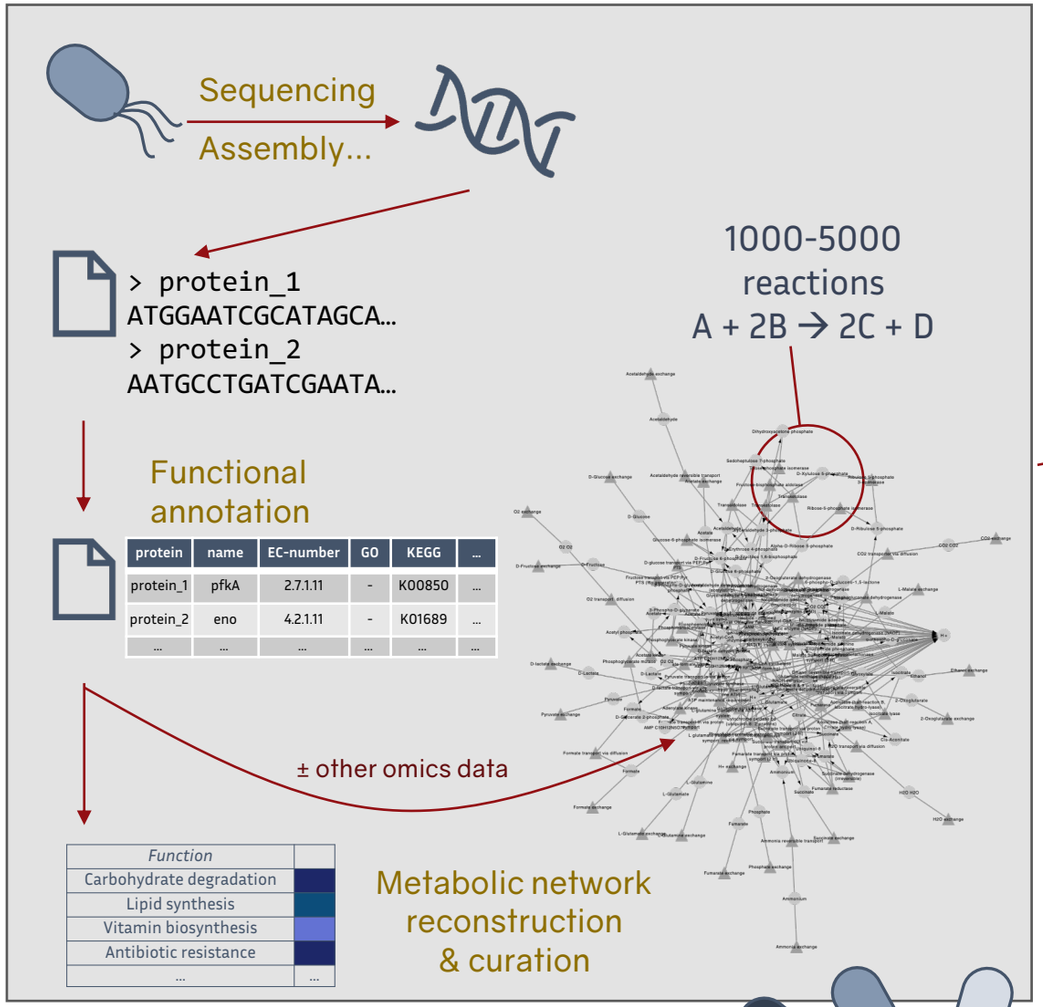
$$S = \begin{pmatrix} +1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & +1 \end{pmatrix} \begin{matrix} M_1 \\ \cdots \\ M_n \end{matrix} \qquad \begin{matrix} r_1 & \cdots & r_n \end{matrix}$$

Automated reasoning using logic paradigms and combinatorial optimisation problems



Picture from Orth et al, Nat. Biot. 2010

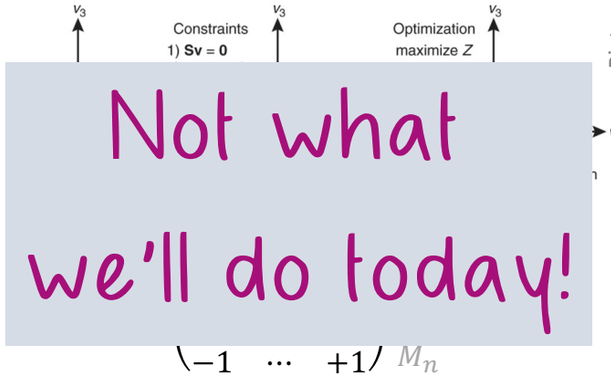
Metabolic modelling



Environmental conditions

± metabolic objective

Flux analysis (FBA, FVA...)
Constraint-based modelling



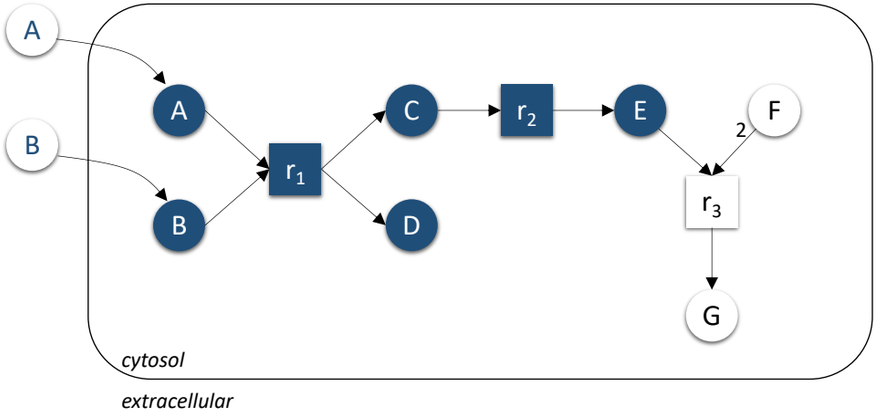
Not what we'll do today!

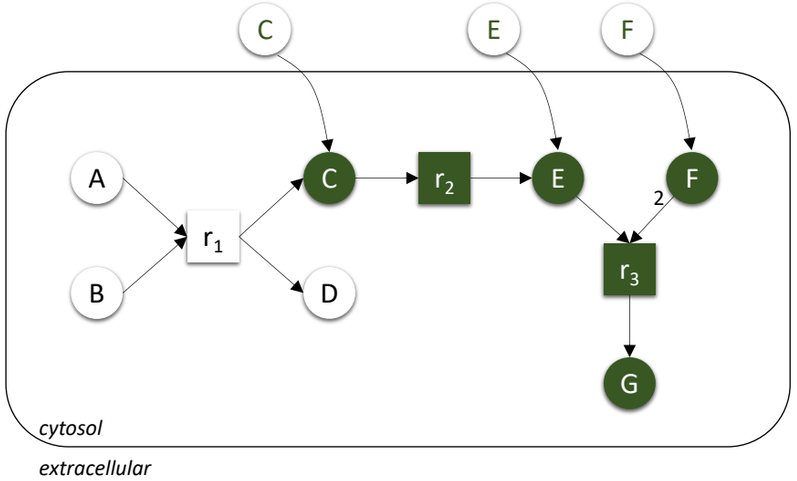
Automated reasoning using logic paradigms and combinatorial optimisation problems

Today's topic!

Picture from Orth et al, Nat. Biot. 2010

Boolean abstraction of metabolic producibility





Network expansion [Ebenhöh et al 2004]

→ Scope of seeds

→ Discrete modelling

$$scope(G,S) = \bigcup_i M_i\,,$$

where $M_0 = S$ and

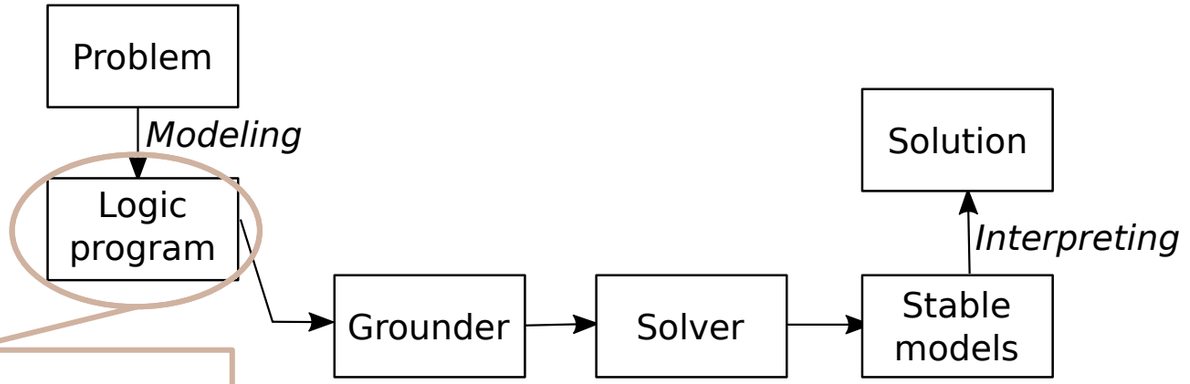
$$M_{i+1} = M_i \cup products(\{r \in R | reactants(r) \subseteq M_i\})$$

Qualitative simulation of metabolic producibility ignoring stoichiometric coefficients and w/o objective function

[Kruse et al 2008] [Handorf et al 2005]

A flexible implementation system

- Answer Set Programming (ASP)
- Logic paradigm
- Knowledge representation & reasoning
- Optimisation + combination to linear prog.

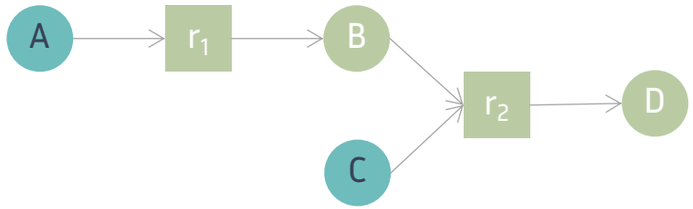



```
reaction(“r1”).
reactant(“A”, “r1”).
product(“B”, “r1”).

reaction(“r2”).
reactant(“B”, “r2”).
reactant(“C”, “r2”).
product(“D”, “r2”).

seed(“A”).
seed(“C”).

scope(M) :- seed(M).
scope(M) :- product(M,R); reaction(R); scope(N) : reactant(N,R).
```

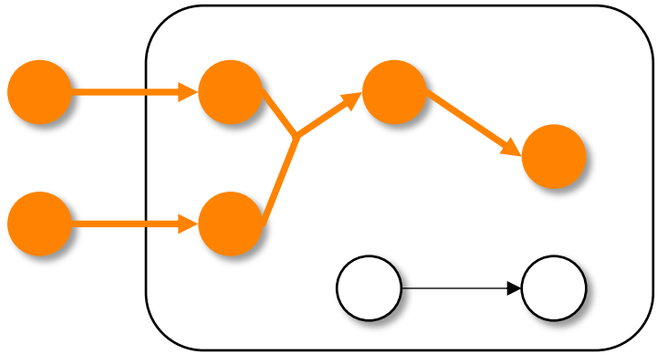


Machine reasoning

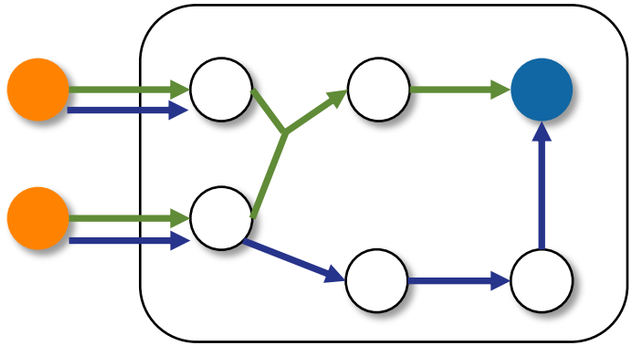
Gebser et al. *Theory and Practice of Logic Programming* (2019)

Applications of the Boolean abstraction of producibility

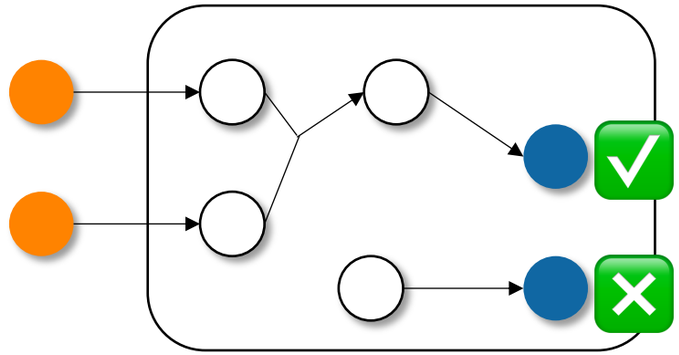
Prigent et al 2017, Aite et al 2017, Frioux et al 2018, Frioux et al 2019, Ghassemi Nedjad et al 20224



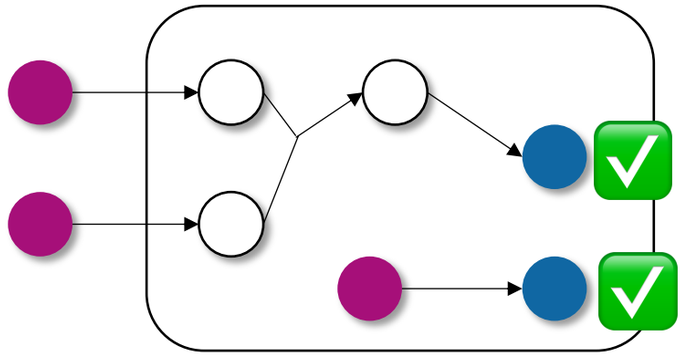
Scope calculation / Activated reactions



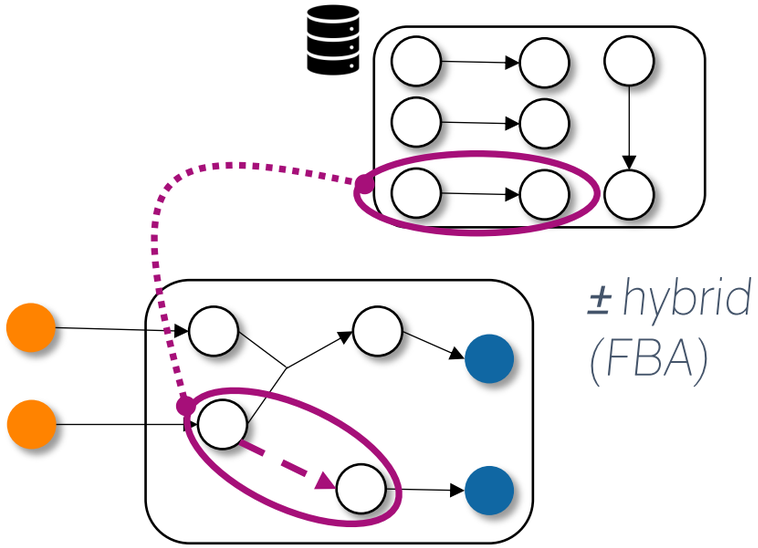
Pathway detection



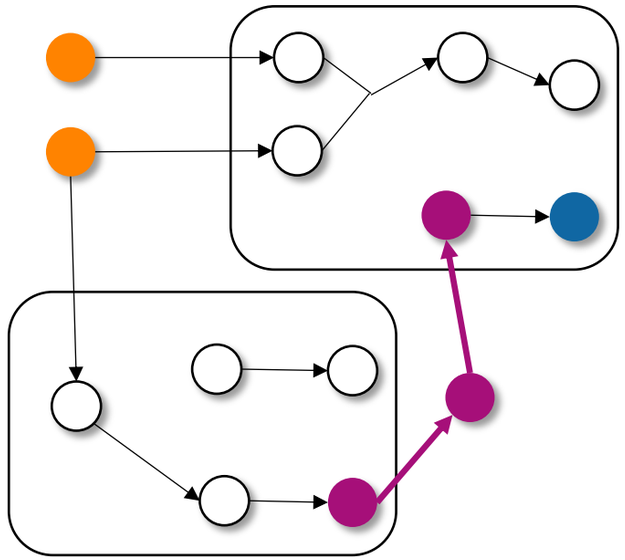
Producibility checking



Medium prediction



Gap-filling

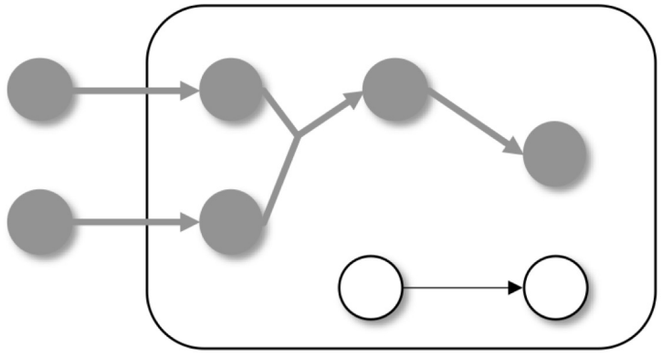


Community analysis

Applications of the Boolean abstraction of producibility

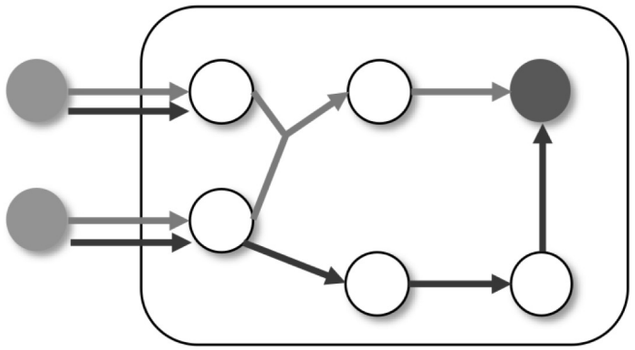
Prigent et al 2017, Aite et al 2017, Frioux et al 2018, Frioux et al 2019, Ghassemi Nedjad et al 20224

MeneTools



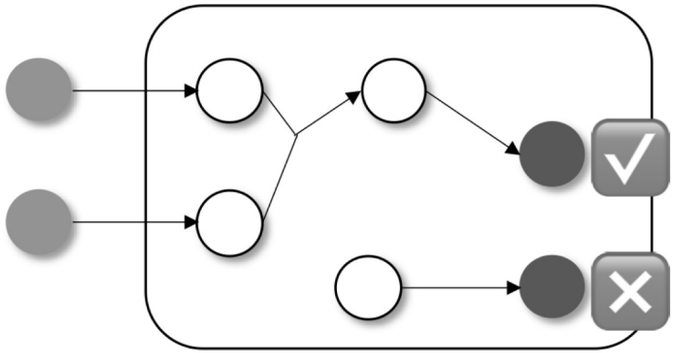
Scope calculation / Activated reactions

MeneTools



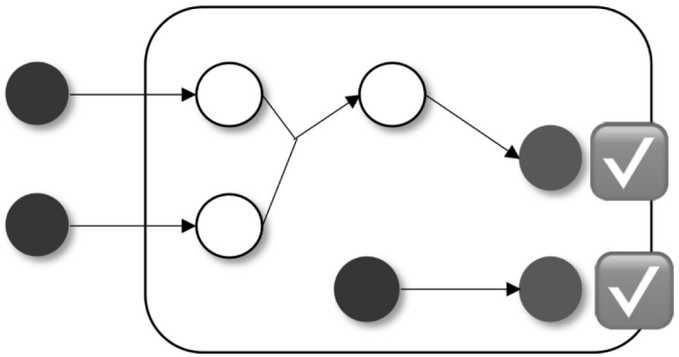
Pathway detection

MeneTools



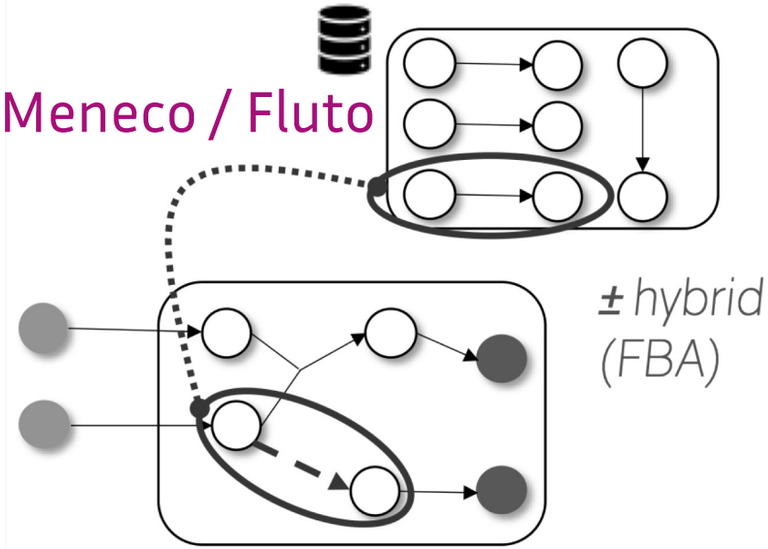
Producibility checking

Seed2LP



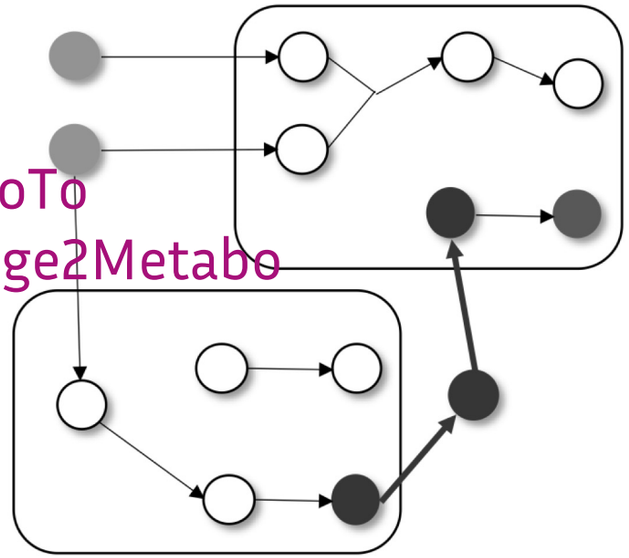
Medium prediction

Meneco / Fluto



Gap-filling

MiSCoTo
Metage2Metabo



Community analysis

Tutorial content

https://gitlab.inria.fr/cfrioux/ebame

1. **Metabolic network exploration:**
    a. Visualisation of metabolic networks
    b. Exploration of the exchange format: SBML
2. **Metabolic network reconstruction**
    a. Structural annotation with Prodigal
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    d. Exploration of metabolic networks reconstructed with Pathway Tools
3. **Metabolic network modelling of a unique species**
    a. Using toy data
    b. Using real data
4. **Metabolic network modelling of a community**
    a. Using toy data
    b. Using real data

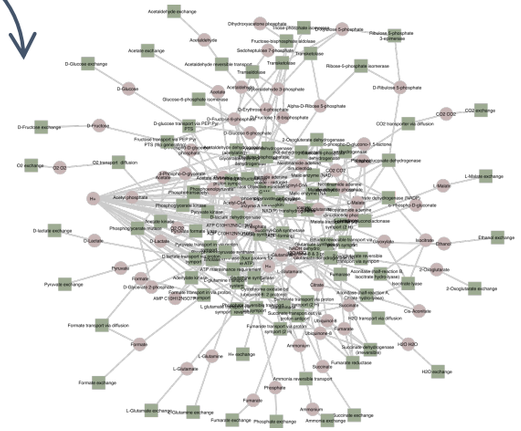
4. Mining metabolic models of microbial communities

Scale, screen, simplify

Metabolic networks, models, and communities



1 genome



1 metabolic network

1,000 to 5,000 reactions

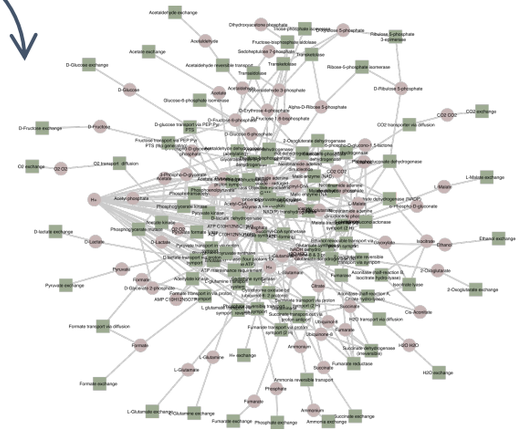
Functional repertoire of the cell.
Can be obtained for all community members through automatic inference

Pacheco et al. *Nat. Comm.* (2019)

Metabolic networks, models, and communities

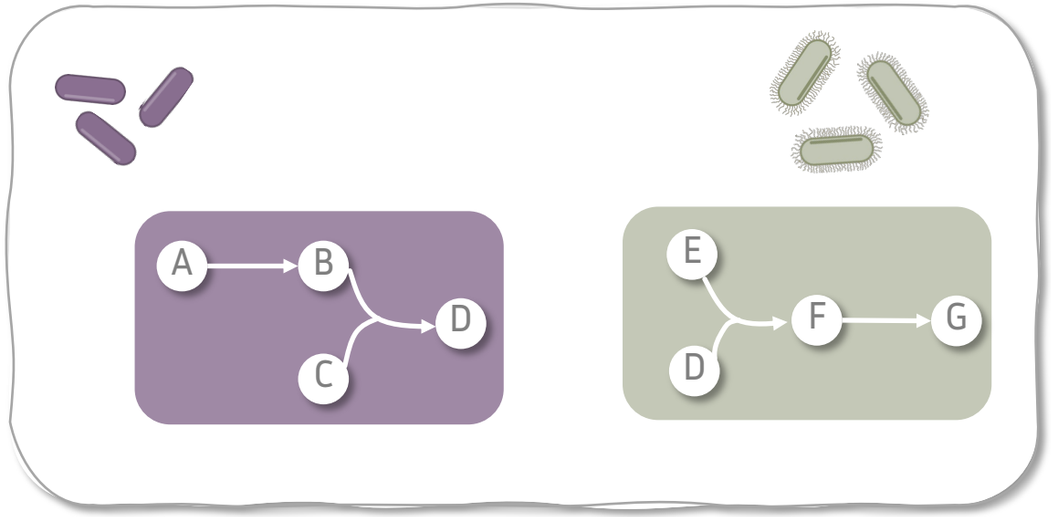


1 genome



1 metabolic network

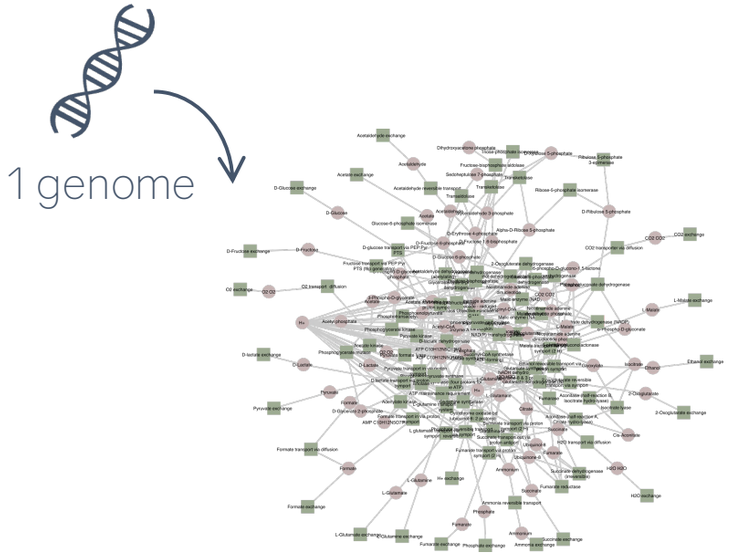
1,000 to 5,000 reactions

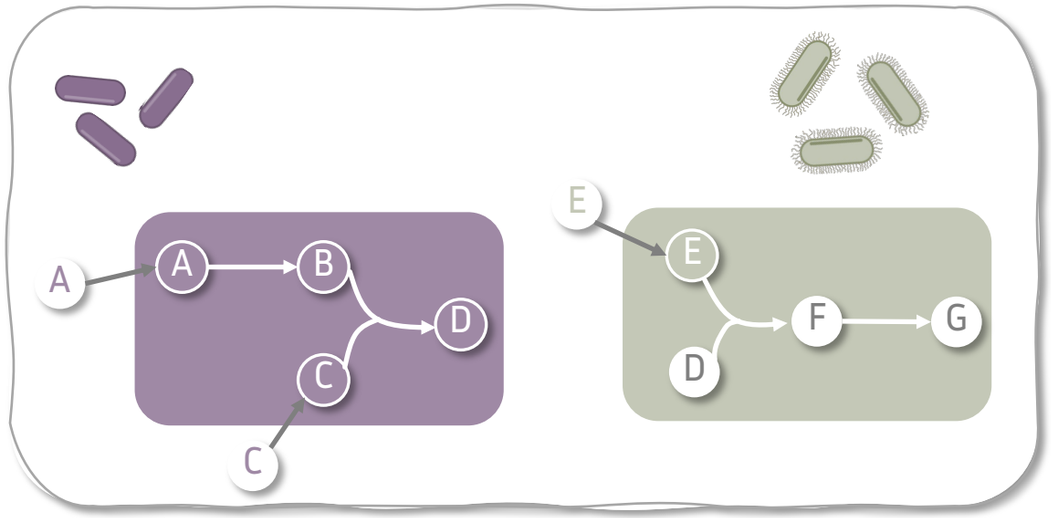


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Pacheco et al. *Nat. Comm.* (2019)

Metabolic networks, models, and communities



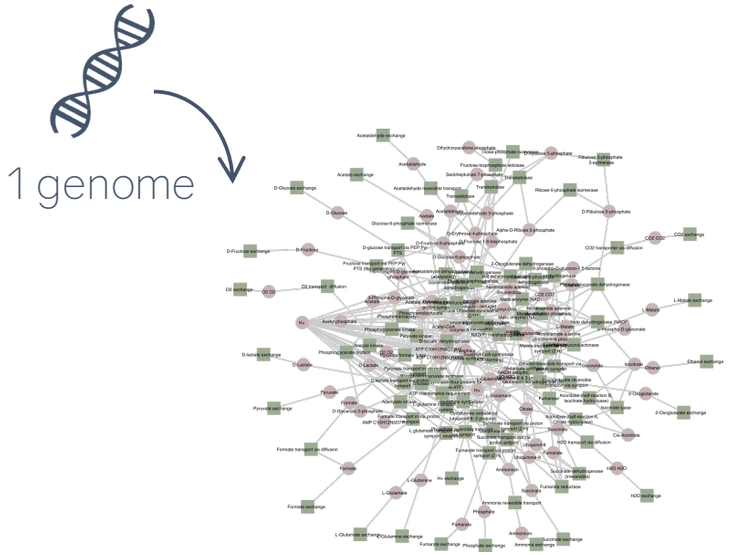


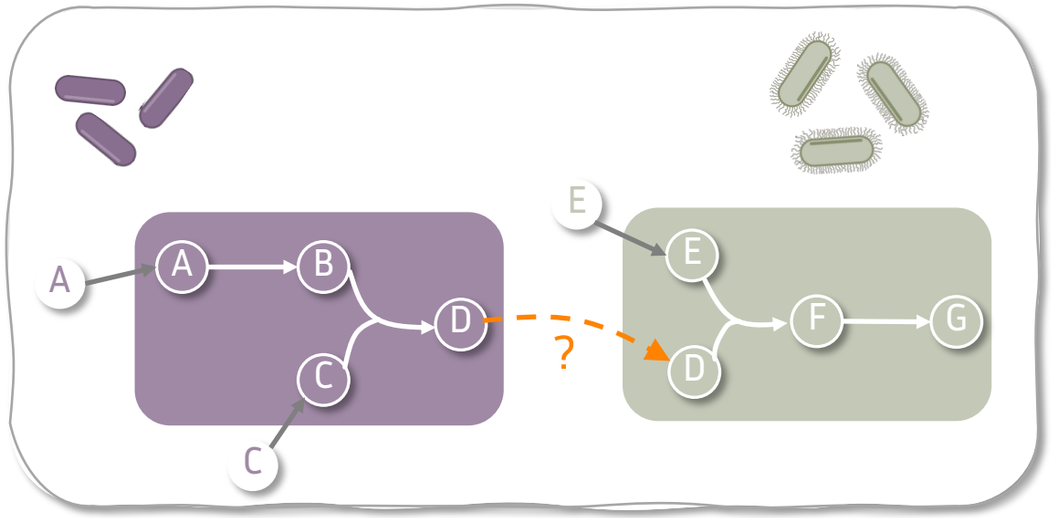
Functional repertoire of the cell.
Can be obtained for all community members through automatic inference

Populations have their individual metabolic potential

Pacheco et al. *Nat. Comm.* (2019)

Metabolic networks, models, and communities





Functional repertoire of the cell.
Can be obtained for all community members through automatic inference

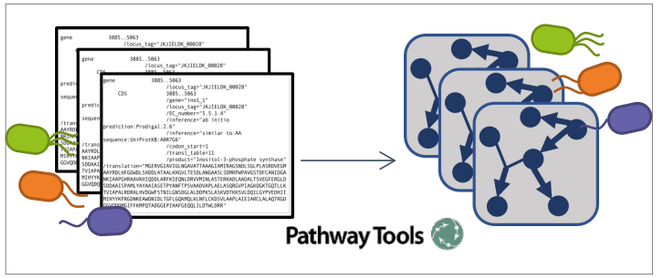
Populations have their individual metabolic potential
Metabolic potential expands through cross feeding in communities, unlocking new metabolic pathways

Pacheco et al. *Nat. Comm.* (2019)

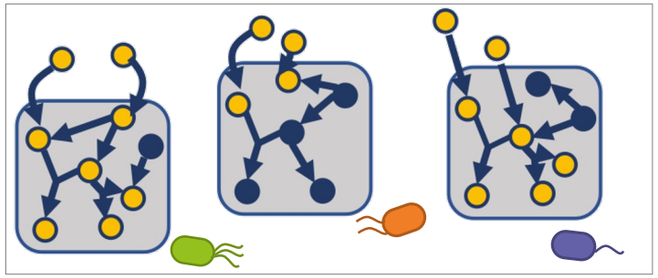
Metage2Metabo:
A pipeline for metabolic screening of communities

Microbiota
- MAGs
- Reference genomes

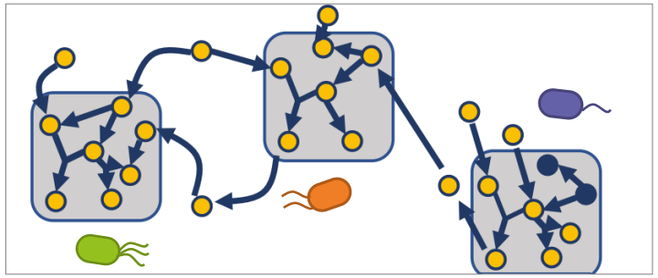
Automatic GSMN reconstruction



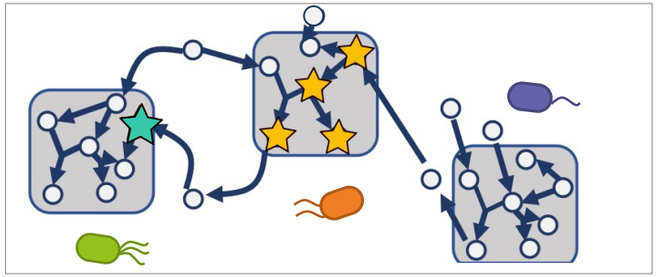
Individual metabolic capabilities



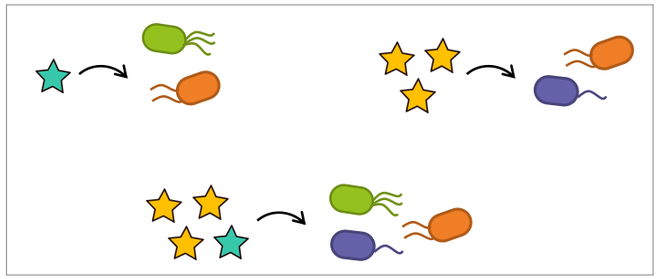
Collective metabolic capabilities



Added value of synergistic interactions



Minimal community selection



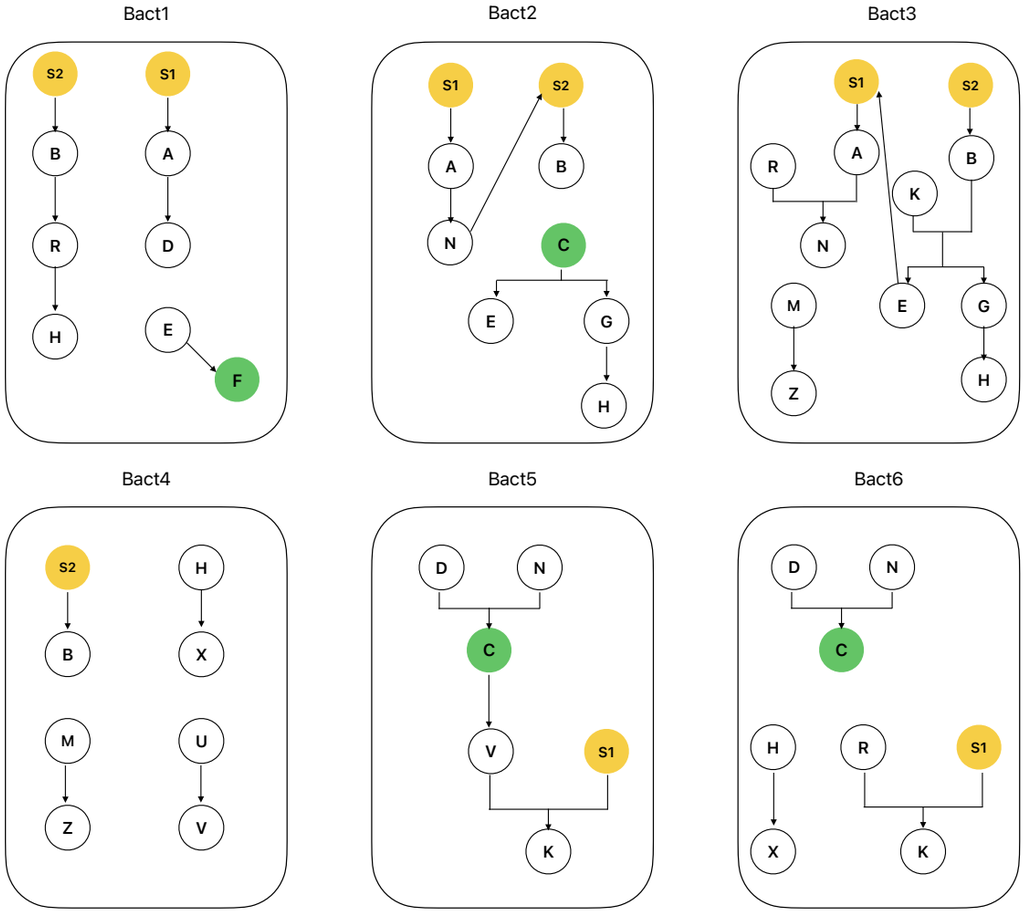
Systematic screening of **metabolic potential** and **mutualistic** potential in a microbiota

`Github`
`Aureme/Metage2Metabo`

[Belcour*, Frioux* et al. eLife 2020]
[Karp et al. Bioinformatics 2002]
[Frioux et al. Bioinformatics 2018]

[Belcour*, Frioux* et al. eLife 2020]

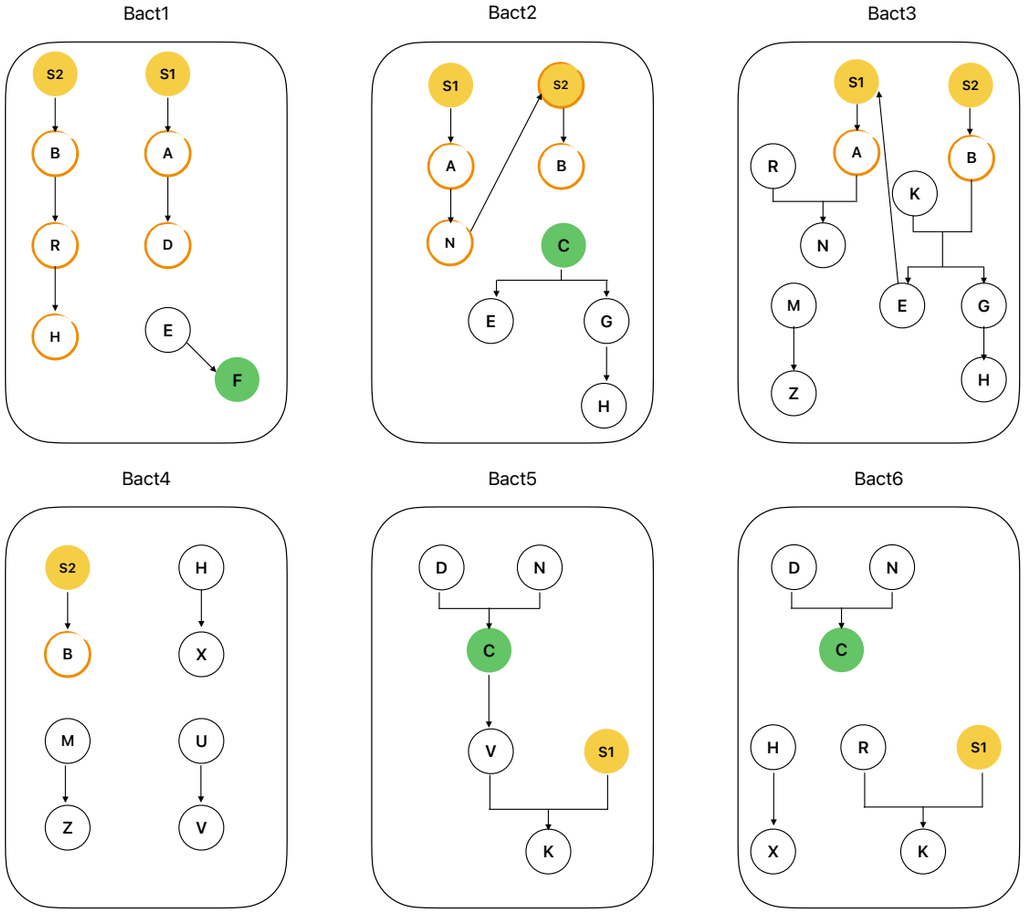
Metage2Metabo: application to an example



A community of 6 species, in an environment (yellow), with compounds of interest (green)

[Belcour*, Frioux* et al. eLife 2020]

Metage2Metabo: application to an example

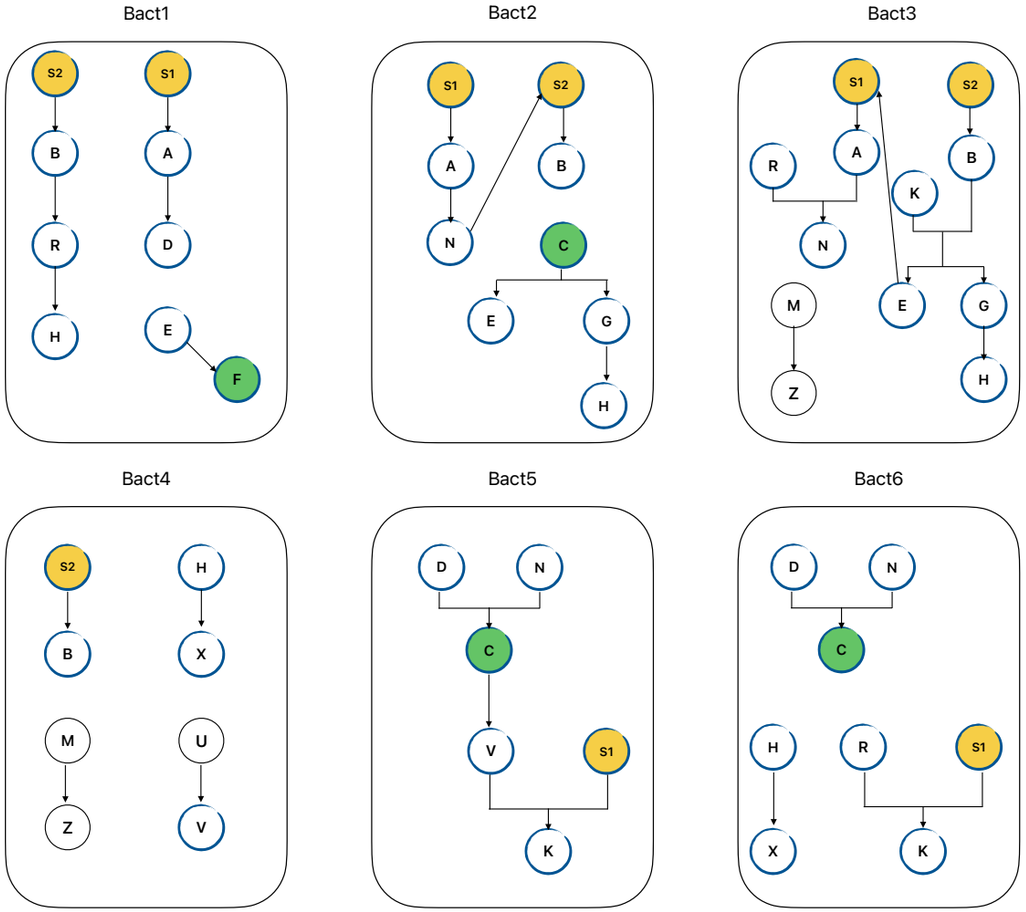


Symbiont	Individual Scope
bact1	A, B, D, R, H
bact2	A, B, N, S2
bact3	A, B
bact4	B
bact5	
bact6	

Individual metabolic potential of each symbiont (orange nodes)

[Belcour*, Frioux* et al. eLife 2020]

Metage2Metabo: application to an example

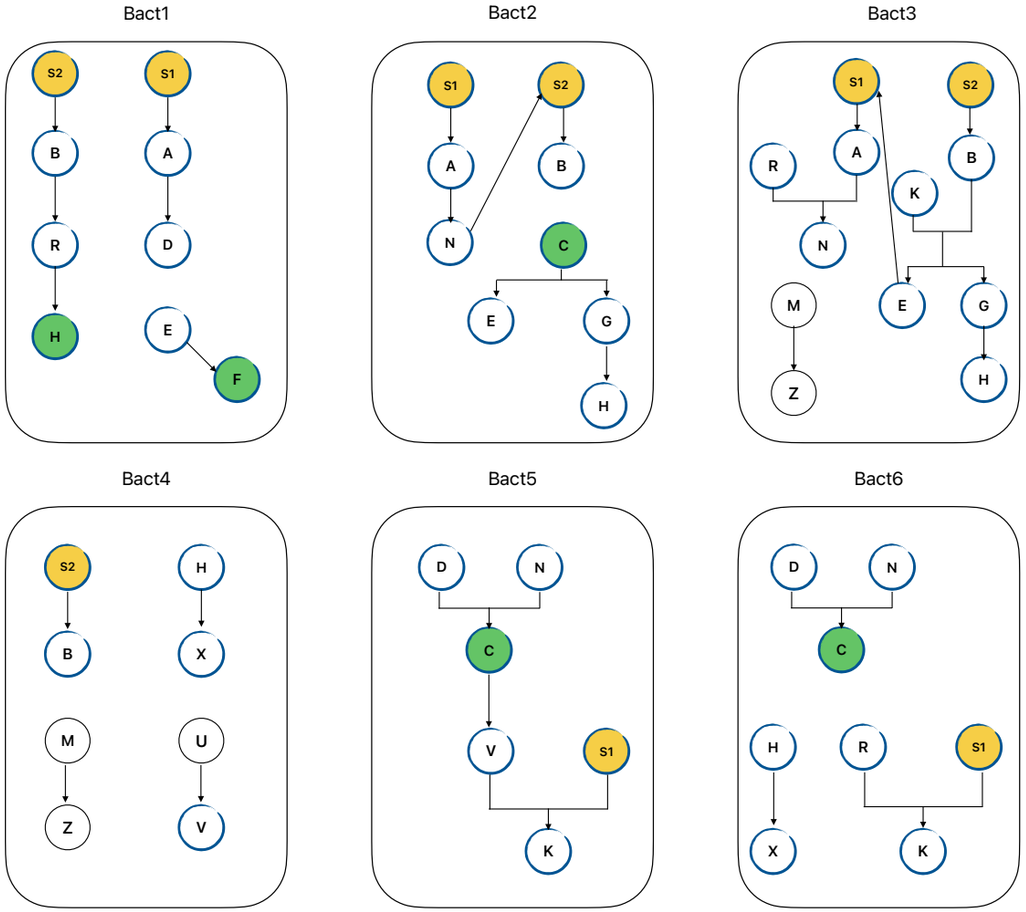


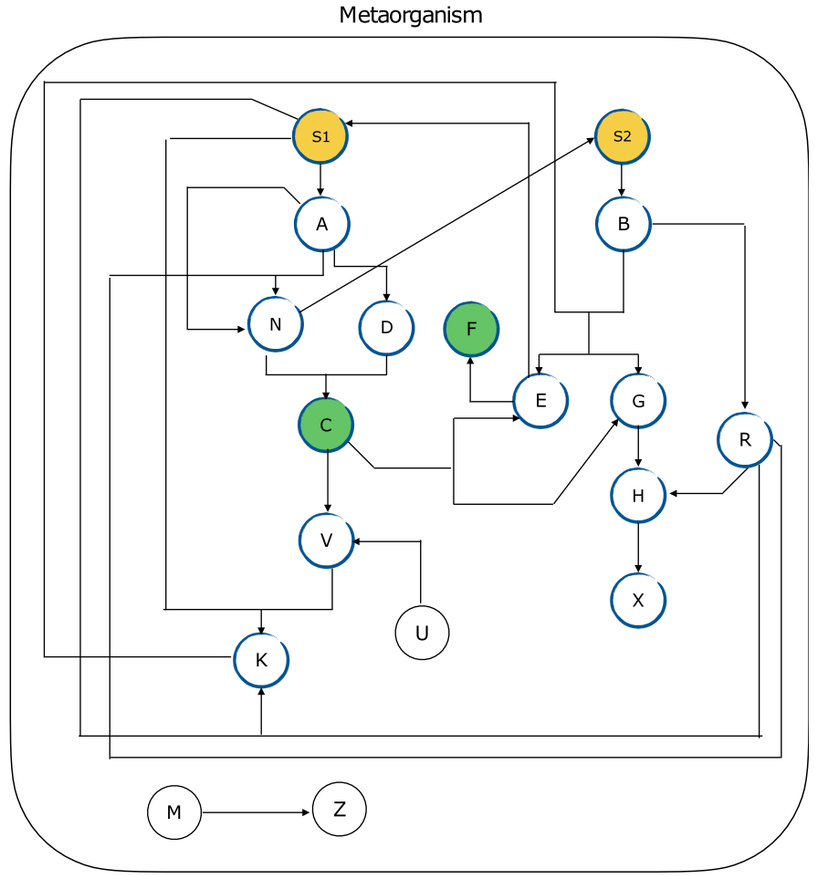
Symbiont	Individual Scope	+ community prod
bact1	A, B, D, R, H	F
bact2	A, B, N, S2	E, G, H
bact3	A, B	E, G, N, H, S1
bact4	B	X
bact5		C, K, V
bact6		C, K, X

Synergistic interactions increase the potential for the community (blue nodes)...

[Belcour*, Frioux* et al. eLife 2020]

Metage2Metabo: application to an example



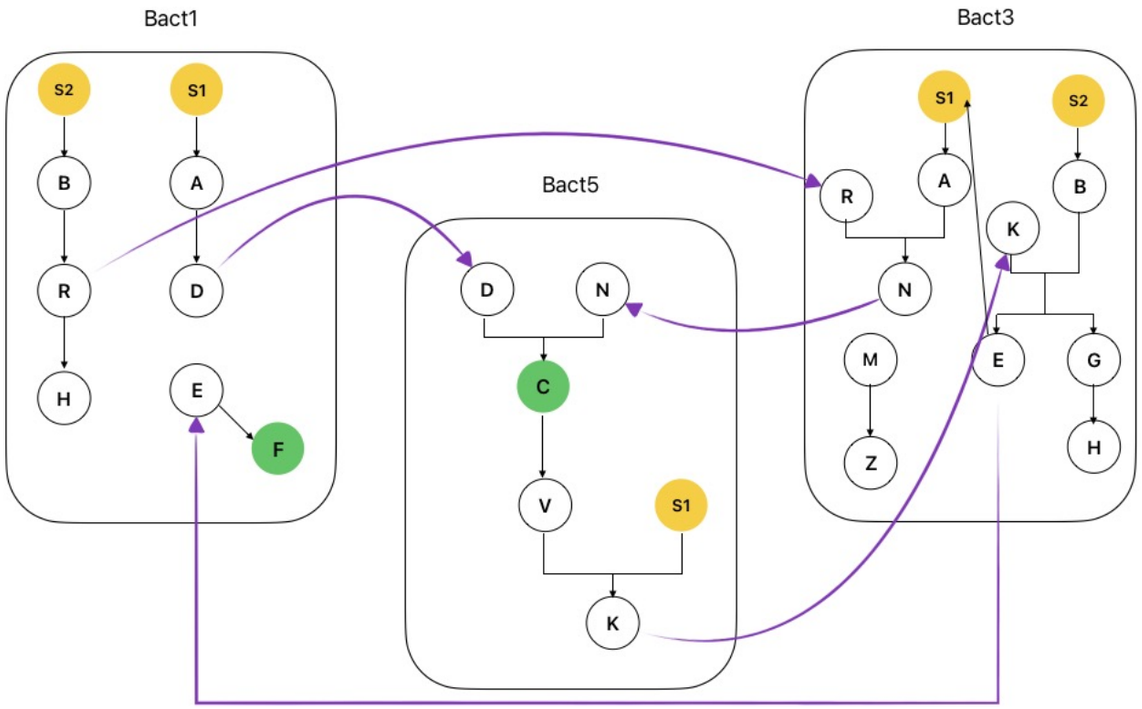


Synergistic interactions increase the potential for the community (blue nodes)…

… similarly to the way a meta-organism of the community would behave

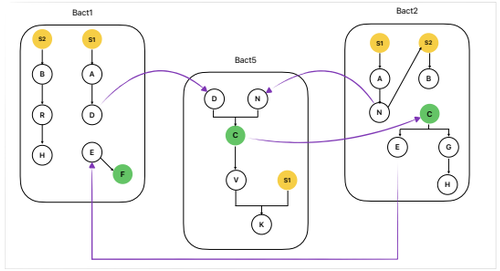
[Belcour*, Frioux* et al. eLife 2020]

Metage2Metabo: application to an example

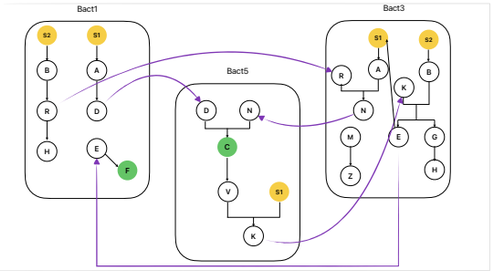


bact1 – bact5 – bact3

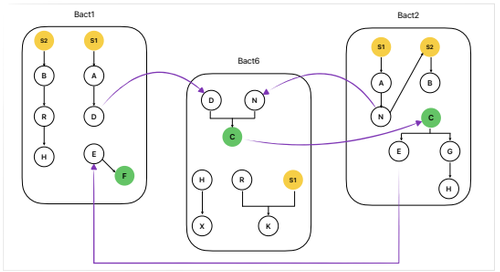
bact1 – bact5 – bact2



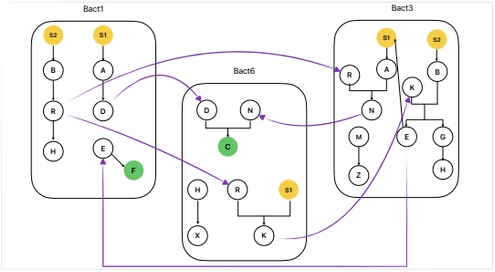
bact1 – bact5 – bact3



bact1 – bact6 – bact2



bact1 – bact6 – bact3



4 equivalent solutions,
4 to 5 metabolic interactions

Selecting a minimal community producing the compounds of interest

Possibility to enumerate all solutions or to get the players occurring in minimal communities

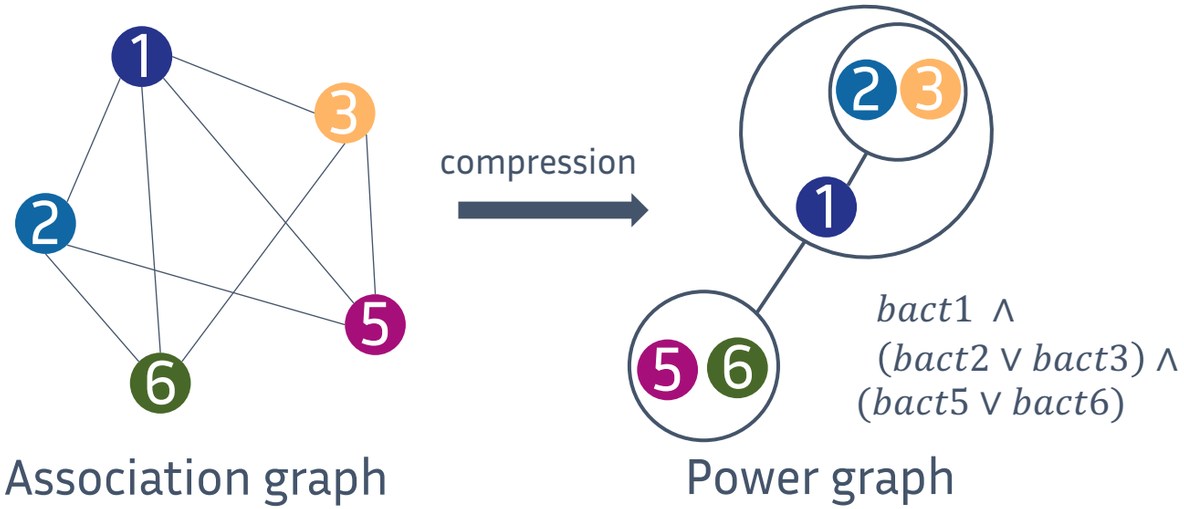
[Belcour*, Frioux* et al. eLife 2020]

Metage2Metabo: application to an example

Minimal communities:
- bact1 - bact2 - bact5
- bact1 - bact2 - bact6
- bact1 - bact3 - bact5
- bact1 - bact3 - bact6

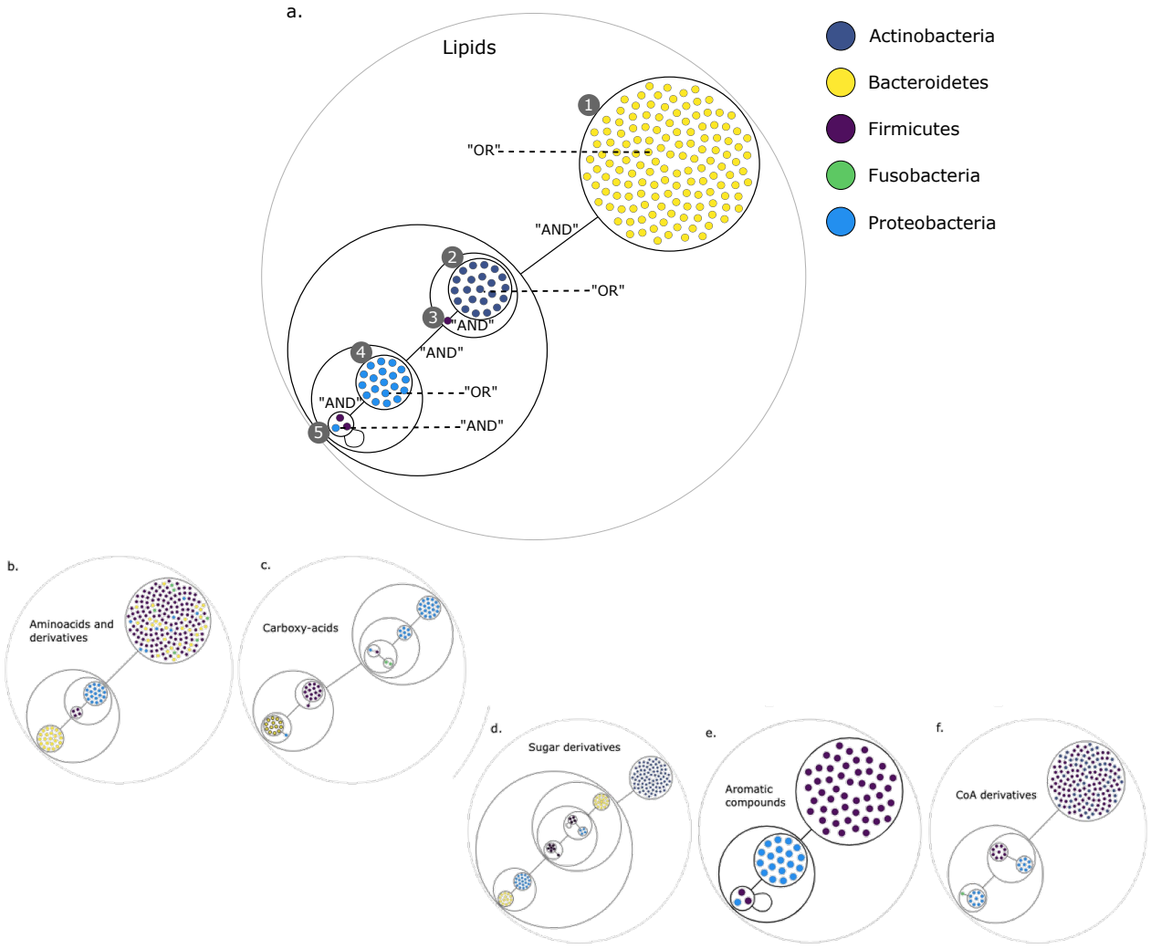
KEY SPECIES





Association graph

Power graph



Highlighting symbionts with equivalent roles

Reveals interesting patterns on real data

Tutorial content

https://gitlab.inria.fr/cfrioux/ebame

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5. Tutorial

https://gitlab.inria.fr/cfrioux/ebame

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Acknowledgements







Chabname Ghassemi Nedjad

Coralie Muller

Daniel Garza